

TripTap

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a.) brief description of your app along with its key requirements

Brief Description:

TripTap is a travel organization app that helps users create and manage trips, store essential travel documents, and view destination weather conditions and time. The app provides a centralized hub for travel.

Key Features and requirements:

- **Trip Management:** Create, edit, and delete trips with destination details, date ranges, and purposes of visit.
- **Dashboard:** Display trips categorized as active, upcoming, or past with visual indicators
- **Document Storage:** Capture and store travel related documents with support for images and PDFs
- **Destination Information:** Display current weather and local time for trip destinations
- **Trip Mapping (added feature):** Visualize trip destinations on a map with appropriate status indicators markers.

b.) Development challenges and interesting design/implementation decisions, if any:

Google Maps Setup

Setting up Google Maps integration: The Markers

Marker Management:

Creating and updating map markers dynamically based on the specific trip status required implementing a custom marker creation system. Where the marker was used based on the current state of the current trip. If the state was active have the marker show as green and show destination, if trip was upcoming show as blue marker and if trip was upcoming show as purple.

Camera Integration for Document Capture:

Challenges: platform permission to allow access to camera in IOS and duplicate image names.

Image with timestamp names to avoid so image gets guarantee unique name when being stored, adding additional error handling for unwanted scenarios.

Setting up Weather Service Logic:

Weather being parse as Fahrenheit issues since default is in C. Fahrenheit must be parsed manually and set up as the current weather. Likewise making sure it is using correct weather city, used a list of cities with coordinated to specify the location.

C.) Technical lessons that you learned, including significant Flutter features/libraries

- The Specific usage of Flutter widgets to best enhance the building of my project idea. The usage of gridview and Listview for a clean and organized way of displaying relative data across the app. Using complementaty widgest such as the slidable, drop down and the interactive zoom which allowed the ability to view documents more efficiently. Thought out my project I learned that during actual implementation design choice change and I had to adapt to them while not changing the project overall and not increasing the complexity.

The project creation of provided me valuable technical knowledge of this cross platform development. The implementation of separating between data models and logic and other UI improved overall maintainability which I Intend and use in the future. I also learned how to use specific APIs and explored a variety which I will

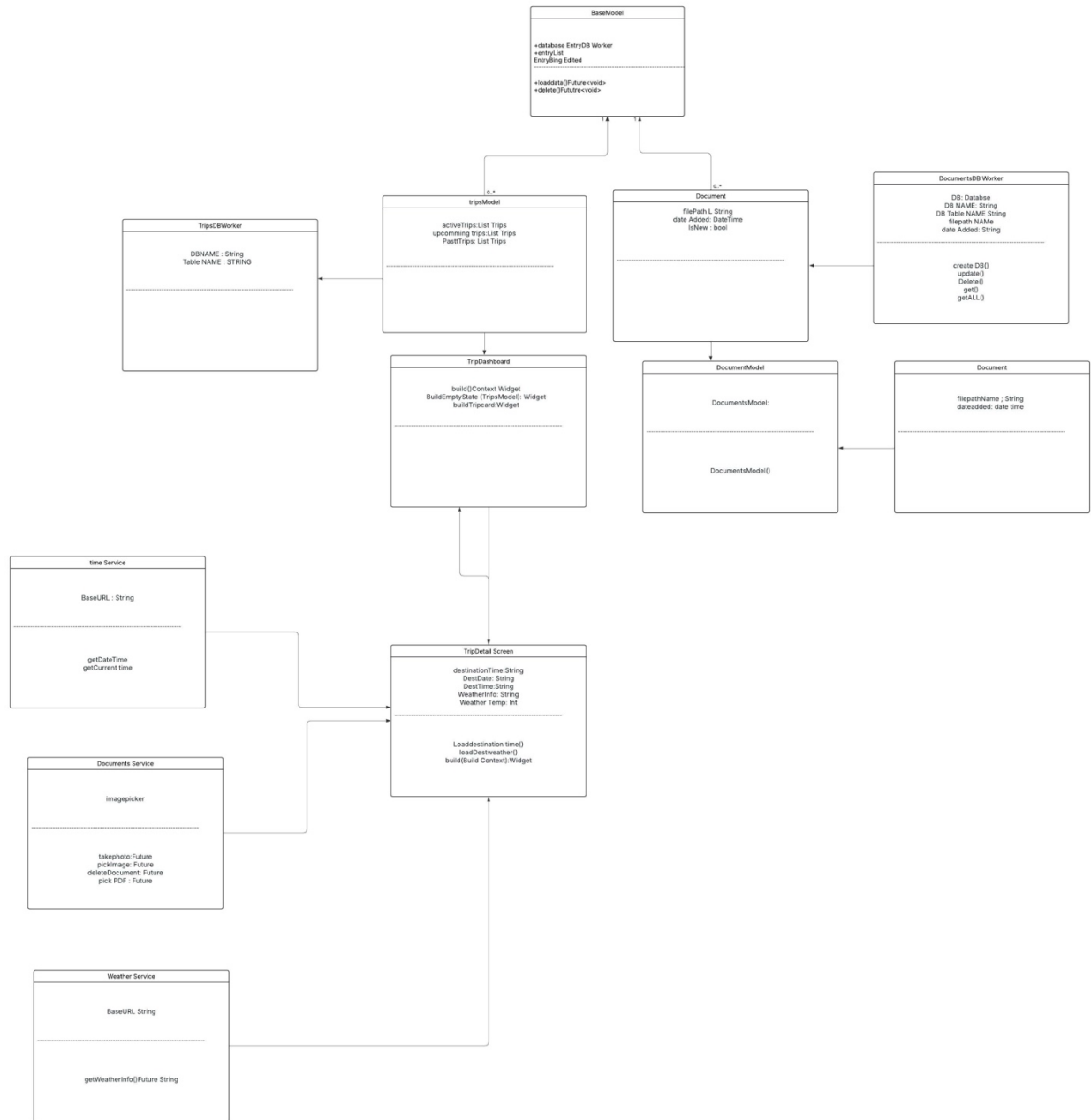
implement in the future for improvement of this app and newer apps that I have planned. In conclusion this project helped me develop my semester skills and also challenged me on researching and implementing my own features in such way that demonstrated my ability to integrate new features on my own.

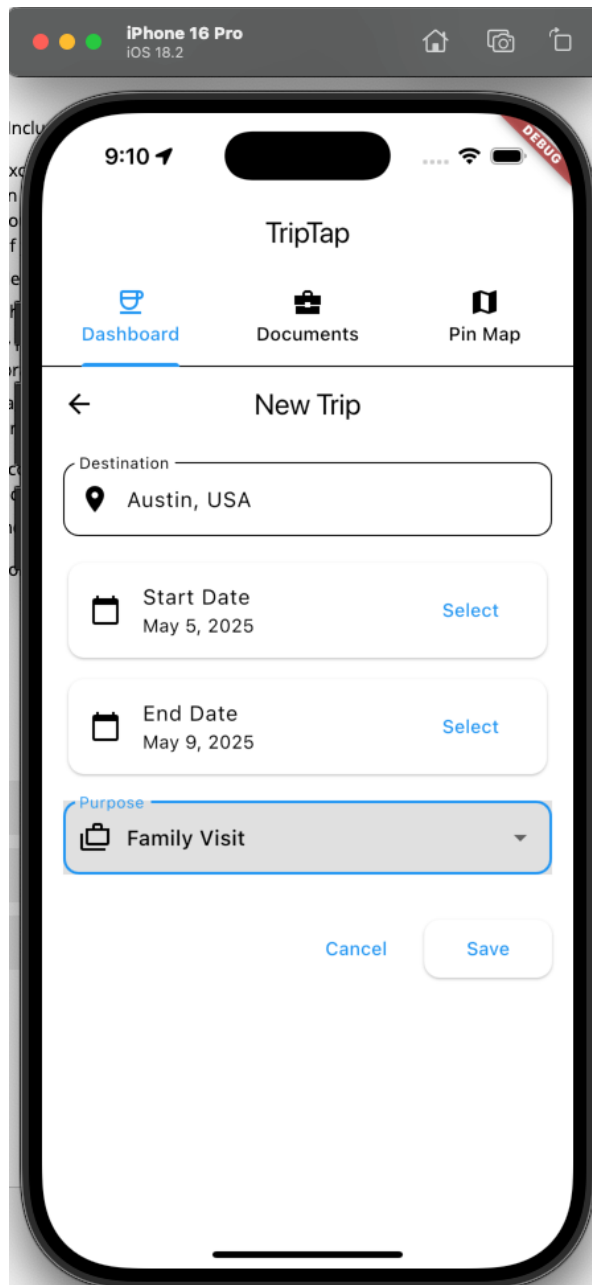
D.) Project complexity in terms of the source code size (number of classes and total lines of code) and the Flutter features/libraries used

15 total files, about 2000 lines of code.

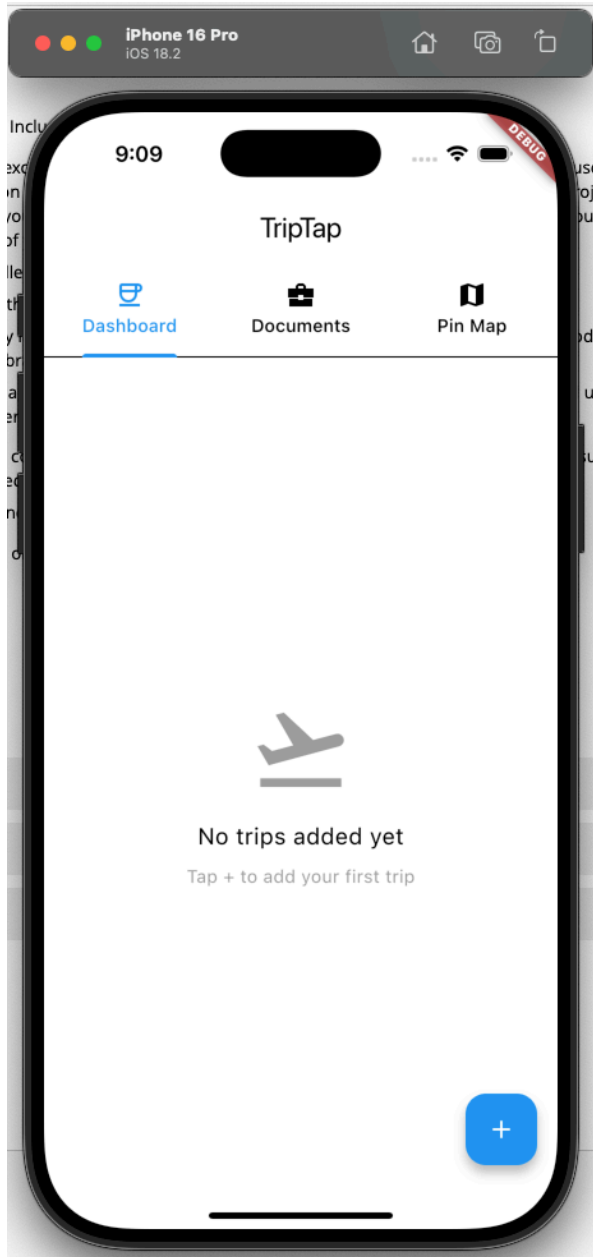
Flutter features/ lib used: Material Design, Form, ScopedModel, Indexed Stack, sqllite for data base, path provider for the file path on system, file picker, flutter slidable, intl for date and time. Flutter pdf viewer. Http, timezone and google maps.

E.) appendix: class diagram and demo screens. Include a series of screens to show the typical use of your app as well as interesting features; provide a brief description for each screen.

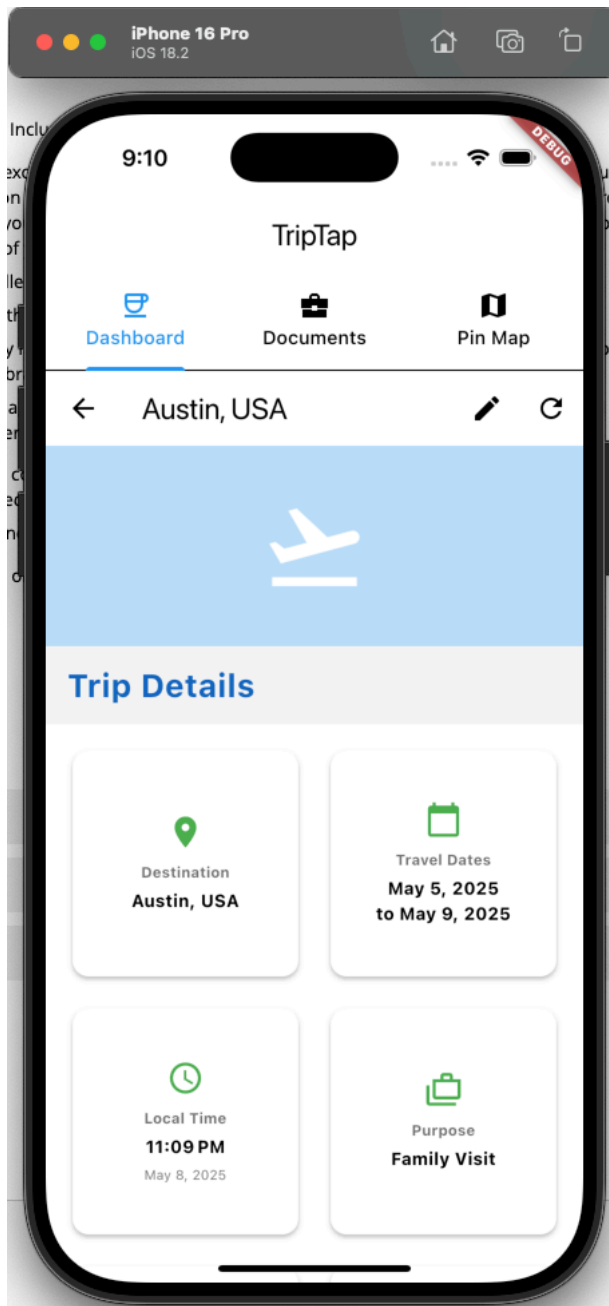




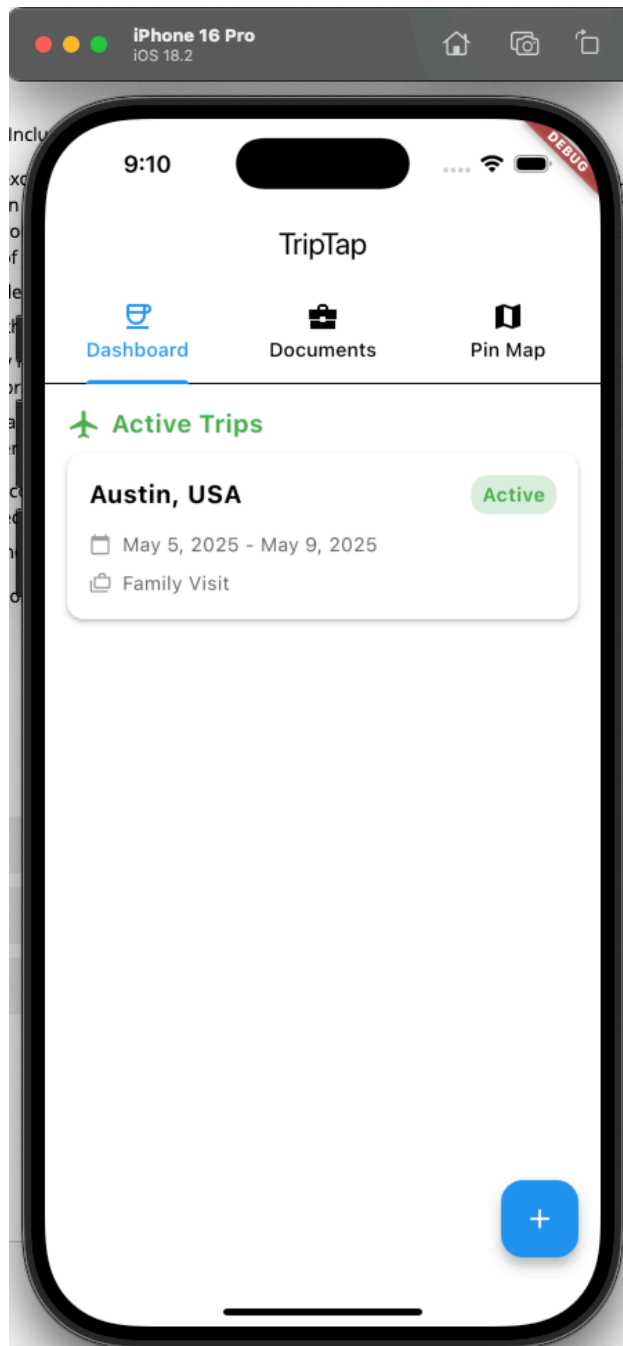
Add new trip screen, destination start and end date and purpose of visit.



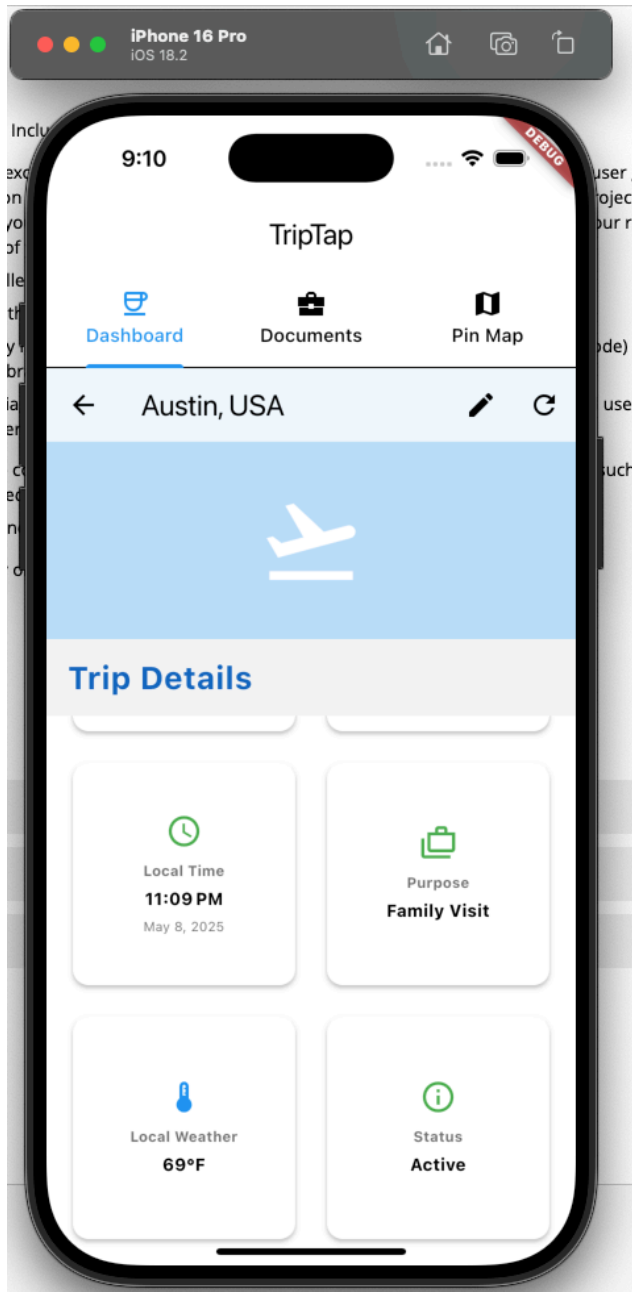
Trip tap dashboard screen empty with no trips.



Trip detail screen showing trip destination, travel dates, local time and purpose of visit.



Trip dashboard screen showing added trip as active



Screen showing detailed trip details, showing local time purpose of visit, local weather and status as active.

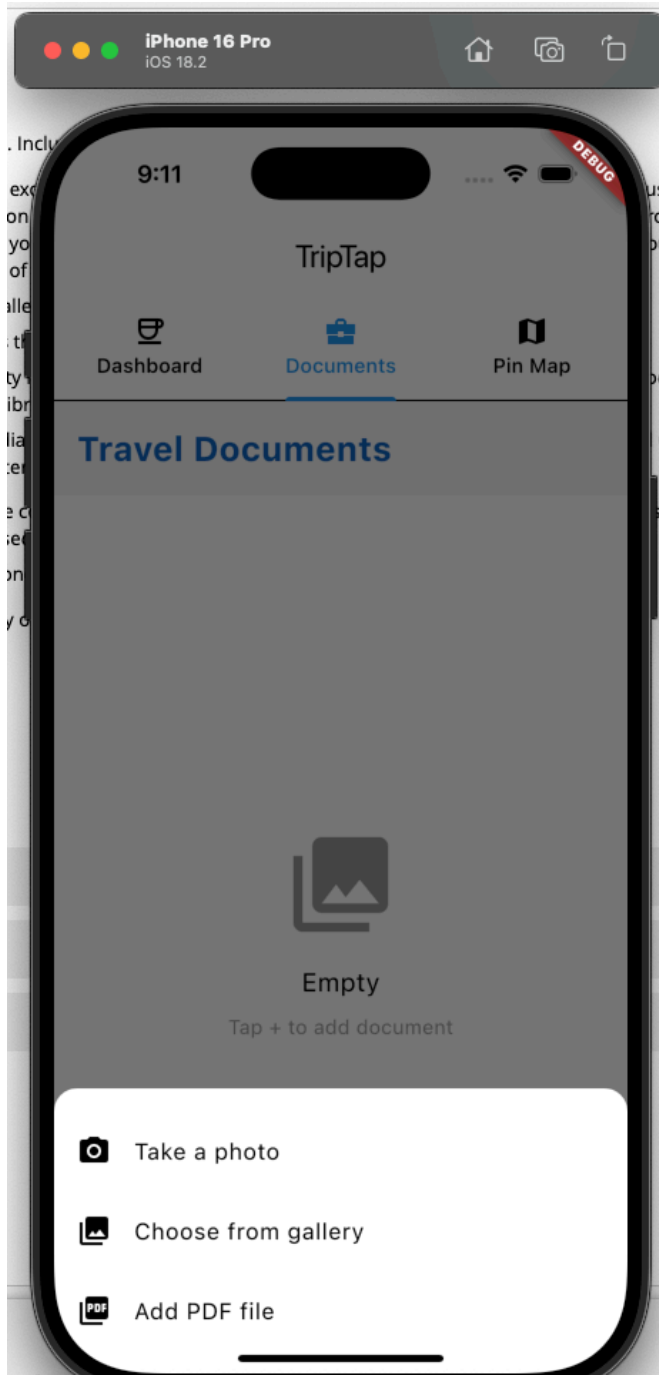
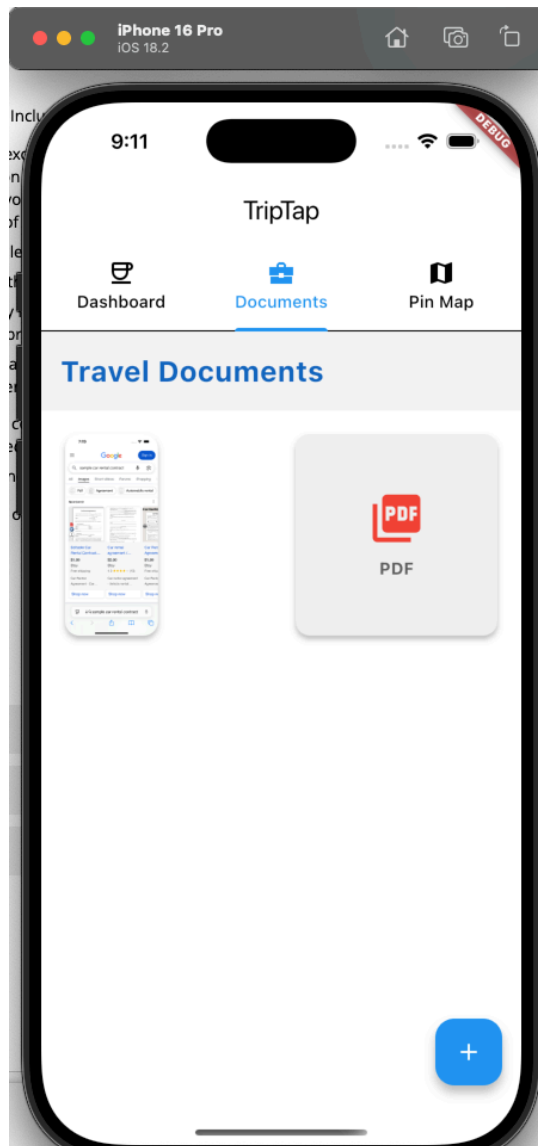
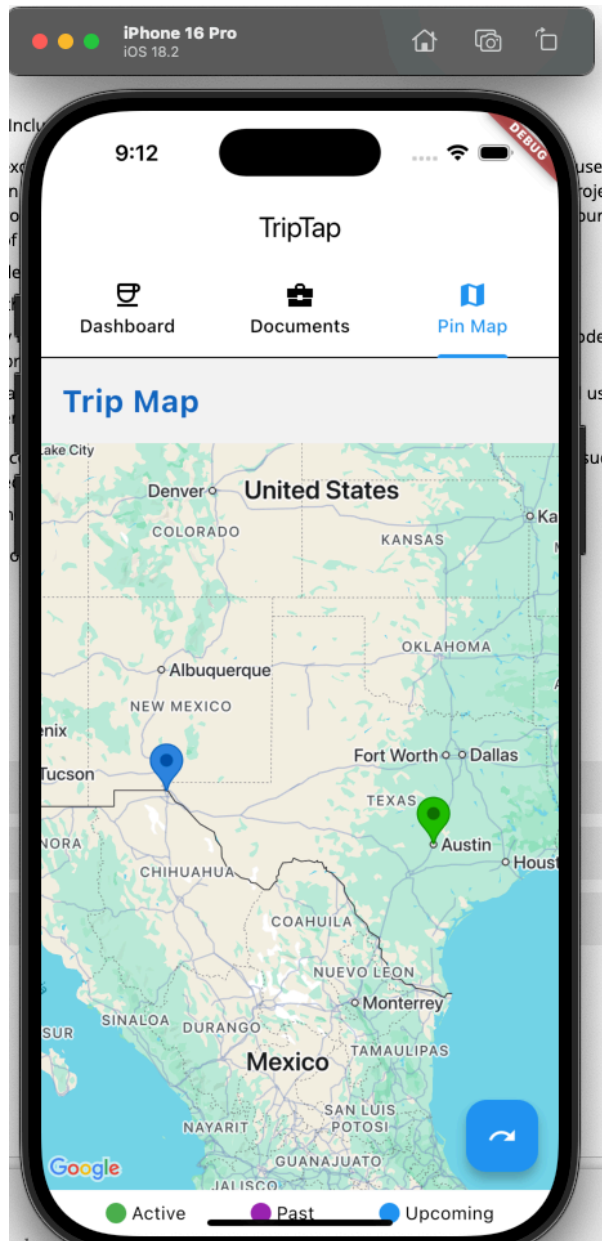


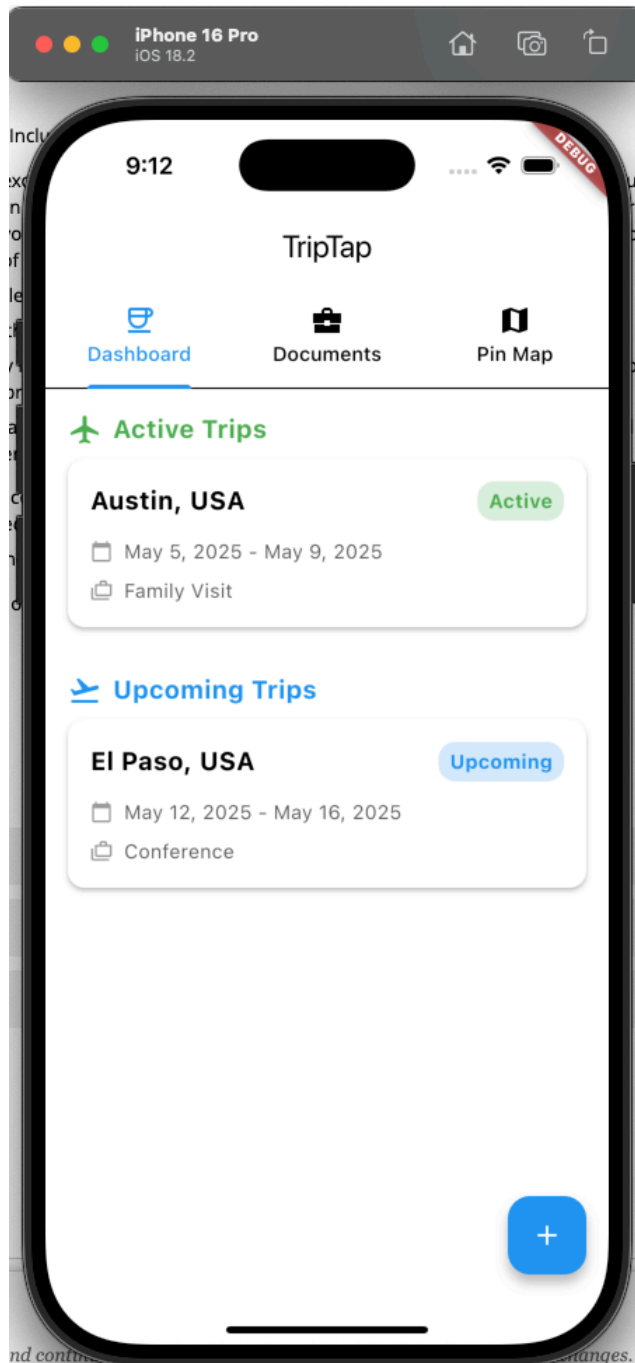
Image showing the ability to add a captured photo using Iphone camera, ability to add photo from gallery, and ability to add PDF file.



Screen showing added both image and PDF.



Screen showing both added trips with different color markers, blue for upcoming , green for active and past will show up as purple.



Screen showing added trips as active and upcoming.